PROJECT REPORT

Touchless HCI for Media Control Using Hand Gestures on NVIDIA Jetson Nano

Submitted by

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Certificate

This is to certify that the project entitled **"Touchless HCI for Media Control Using Hand Gestures on NVIDIA Jetson Nano"** is a bonafide work carried out by Kowshik Repaka, Potnuru Karthik, and Yennapusa Gayathri under the guidance of Dr. B V V Satyanarayana, Department of Electronics and Communication Engineering, Vishnu Institute of Technology, during the academic year 2025-2026. The work is original and carried out in partial fulfillment of the requirements for the award of the degree.

Project Guide: **Head of the Department:**

Abstract

This project presents a comprehensive design and implementation of a touchless Human-Computer Interaction (HCI) system for controlling multimedia applications using hand gestures on the NVIDIA Jetson Nano platform. The increasing adoption of smart devices and interactive systems has created a demand for natural, hygienic, and hands-free interaction. Traditional input methods require physical contact and are inconvenient in shared environments.

The proposed system captures real-time video using a USB camera. OpenCV is used for preprocessing and MediaPipe for hand landmark detection. Gestures such as closed palm (pause), one finger (play), two fingers (volume up), three fingers (volume down), four fingers (forward), and open palm (backward) are recognized. PyAutoGUI automates keyboard shortcuts for media control. The Jetson Nano enables low-latency edge processing without cloud dependency. SimpleScreenRecorder is used for documentation and evaluation. Results demonstrate reliable real-time performance in indoor environments. Future work includes multi-user interaction and robustness improvements.

Contents

Problem Statement

Despite the widespread availability of multimedia systems, user interaction with these systems still largely depends on physical input devices or voice commands. Physical controllers require contact, which may be inconvenient in situations involving hygiene concerns, shared devices, or users with limited mobility. Voice-based control, while useful, may fail in noisy environments, raise privacy concerns, and be unsuitable in quiet settings such as libraries or classrooms.

The problem addressed in this project is the absence of a reliable, low-cost, and real-time touchless interface for controlling media playback in everyday environments. The system should be capable of accurately recognizing hand gestures using a standard camera and mapping them to common media control actions with minimal latency. Additionally, the solution should be deployable on an embedded platform, operate without continuous internet connectivity, and be user-friendly for non-technical users.

Introduction

Human-Computer Interaction (HCI) is an interdisciplinary field that focuses on the design, evaluation, and implementation of interactive computing systems for human use. Traditional interaction mechanisms such as keyboards, mice, touch screens, and remote controls have dominated personal computing for decades. While these interfaces are effective, they impose limitations in scenarios where hands-free interaction is required, physical contact is undesirable, or users experience motor impairments.

Recent advancements in computer vision, machine learning, and embedded computing have enabled new forms of natural user interfaces. Gesture-based interaction allows users to communicate with machines using intuitive hand movements, closely resembling human-to-human communication. Such interfaces are particularly valuable in smart homes, interactive kiosks, healthcare environments, and public spaces where hygiene and accessibility are critical concerns.

The NVIDIA Jetson Nano is a low-cost embedded AI platform capable of running real-time deep learning and computer vision applications at the edge. By leveraging MediaPipe for hand landmark detection and OpenCV for image processing, it is possible to implement robust gesture recognition systems on resource-constrained devices. This project explores the feasibility of deploying a practical touchless HCI system for media control on the Jetson Nano, demonstrating how embedded AI platforms can support real-time, intuitive user interaction without relying on cloud-based processing.

Objectives

The main objectives of this project are as follows:

- To design and develop a touchless Human-Computer Interaction (HCI) system for media control using hand gestures.
- To implement real-time hand detection and landmark extraction using computer vision techniques.
- To accurately classify hand gestures based on finger count and palm posture.
- To map recognized gestures to multimedia control functions such as play, pause, next, previous, and volume control.
- To achieve low-latency performance on the NVIDIA Jetson Nano using edge-based processing.
- To ensure hygienic and hands-free interaction suitable for public and assistive environments.
- To evaluate system performance in terms of accuracy, response time, and stability under normal indoor conditions.
- To provide scope for future enhancements such as deep learning-based gesture recognition, multi-user interaction, and multimodal interfaces.

Literature Review

Early research in hand gesture recognition relied on color-based segmentation techniques and handcrafted features such as contours, convex hulls, and shape descriptors. These methods were highly sensitive to lighting variations, background clutter, and skin color diversity. Subsequent approaches employed machine learning classifiers trained on engineered features, improving robustness but still facing challenges in real-world environments.

Recent advancements leverage deep learning and landmark-based detection methods. MediaPipe provides a lightweight framework capable of real-time hand landmark detection on mobile and embedded platforms. Studies have demonstrated the application of gesture-based interfaces in smart home control, robotics teleoperation, and assistive technologies for individuals with disabilities. Research on embedded AI platforms highlights the feasibility of deploying real-time vision applications at the edge, reducing latency and preserving user privacy by avoiding cloud-based processing.

The reviewed literature indicates that combining efficient landmark detection frameworks with embedded GPU acceleration can yield practical gesture-based HCI systems suitable for real-world deployment.

System Architecture

The system architecture is designed as a modular pipeline consisting of video acquisition, preprocessing, hand landmark detection, gesture classification, and action execution modules. The USB camera captures continuous video frames, which are preprocessed using OpenCV to resize and normalize input frames. Media Pipe processes each frame to detect and track 21 hand landmarks, providing precise spatial information about finger joints and palm orientation.

The gesture classification module analyzes relative landmark positions to determine finger states and palm configurations. Each recognized gesture is mapped to a predefined media control action. PyAutoGUI acts as the automation interface, translating high-level commands into keyboard events that control the media player. The NVIDIA Jetson Nano serves as the processing unit, utilizing GPU acceleration to ensure real-time performance. The modular architecture facilitates scalability and future integration with additional interaction modalities such as voice commands or facial gestures.

Hardware Requirements

- NVIDIA Jetson Nano Developer Kit
- USB Camera/Webcam
- microSD Card (32GB+)
- Power Adapter (SV, 4A)
- HDMI Display
- Keyboard
- Mouse
- wifi connection/ethernet

Software Requirements

- Ubuntu (Jetson OS)
- Python
- OpenCV
- MediaPipe
- PyAutoGUI
- SimpleScreenRecorder

Methodology

The methodology follows a real-time processing pipeline. Video frames are captured from the camera and preprocessed to enhance detection stability. MediaPipe is used to extract hand landmarks in each frame. Finger states are computed by comparing relative landmark positions, enabling gesture classification. A decision logic module maps gestures to specific actions such as play, pause, next, previous, volume up, and volume down. PyAutoGUI executes the mapped actions by simulating keyboard shortcuts recognized by the media player.

The system is optimized to minimize latency by using efficient data structures and leveraging GPU acceleration on the Jetson Nano. SimpleScreenRecorder is used to capture execution traces and validate system behavior during testing. Performance evaluation includes qualitative assessment of responsiveness and robustness under varying lighting conditions.

Gesture Mapping

Table 9.1: Gesture-to-Action Mapping

S.No	Gesture	Mapped Action
1	Closed Palm (Fist}	Pause Media
2	One Finger	Play Media
3	Two Fingers	Volume Up
4	Three Fingers	Volume Down
5	Four Fingers	Forward
6	Open Palm	Backward

System Flowchart

Start

Capture Frame

Preprocess (OpenCV)

Landmarks (MediaPipe)

Gesture Classification

Map to Action

Execute (PyAutoGUI)

Next Frame

Figure 10.1: Flowchart of the Proposed System

The flowchart illustrates the real-time workflow of the touchless HCI system on the Jetson Nano. The system initializes required modules and continuously captures hand images using a USB camera. Each frame is preprocessed with OpenCV and analyzed by MediaPipe to extract hand landmarks. Gestures are classified based on finger count and palm posture, mapped to media control commands, and executed using PyAutoGUI. This loop runs continuously, enabling smooth, low-latency, touchless control of multimedia applications.

System Block Diagram

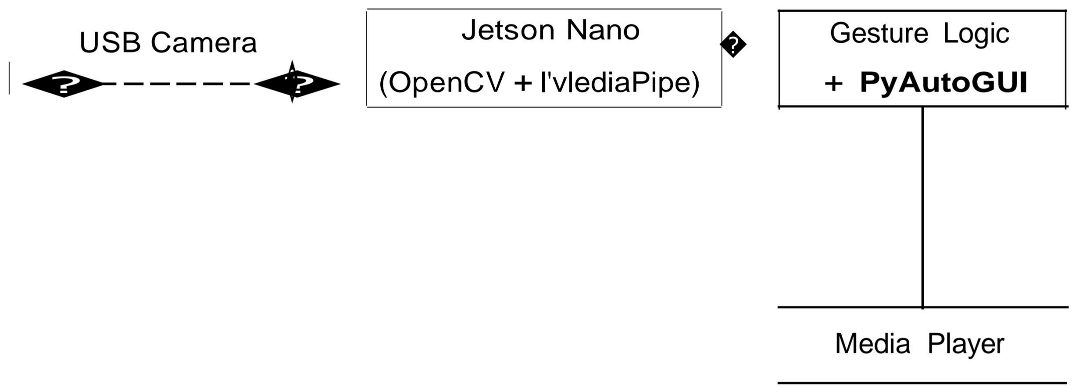


Figure 11.1: Block Diagram of the Proposed System

The block diagram represents the modular architecture of the system, where a USB camera captures hand gestures, the Jetson Nano processes frames and recognizes gestures using OpenCV and MediaPipe, and PyAutoGUI sends the corresponding control commands to the media player. This modular design enables low-latency edge processing and supports future enhancements such as deep learning-based gesture recognition and multimodal interaction.

Testing and Validation

Tests were conducted under varying indoor lighting and backgrounds with multiple users. Validation criteria included correct gesture recognition, low latency, and stable continuous operation. Results recorded using SimpleScreenRecorder confirmed consistent behavior in controlled environments.

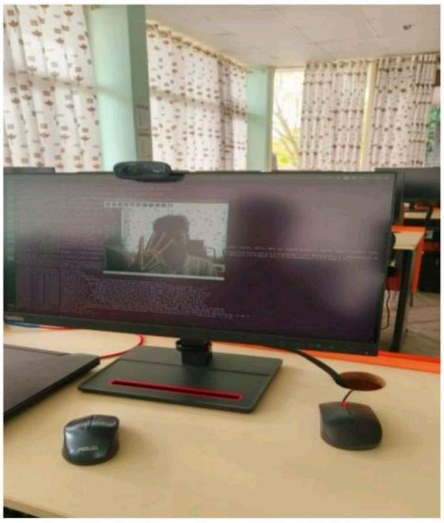
Performance Evaluation

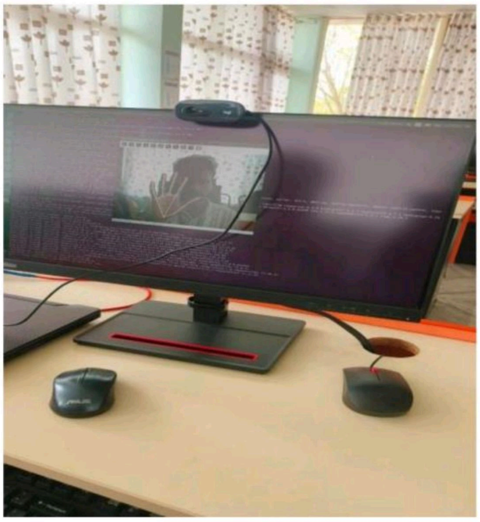
Table 13.1: Accuracy and Performance Metrics

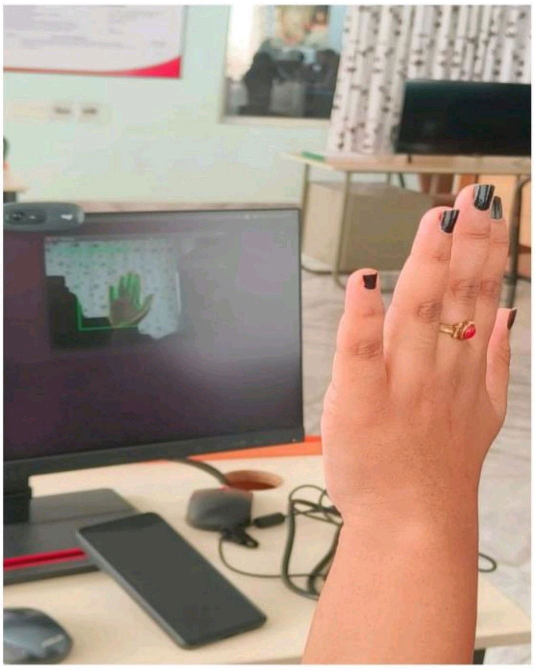
Metric	Observed Value
Gesture Recognition Accuracy	92-95%
Average FPS (Jetson Nano)	18-22 FPS
End-to-End Latency	120-180 ms
False Positives	2-5%

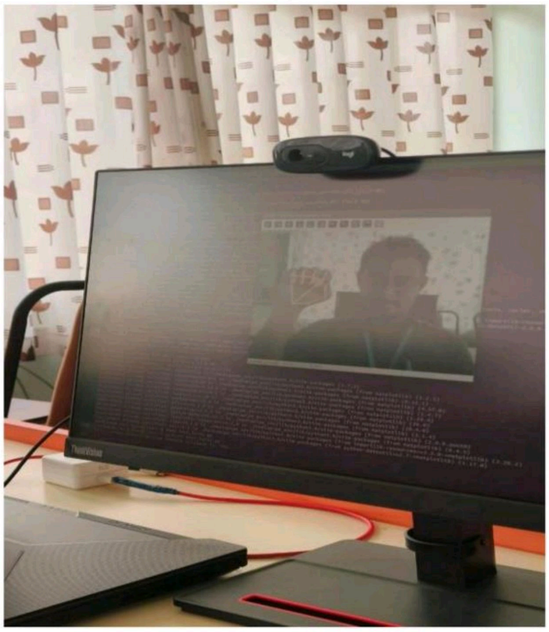
Photos and Results

Photos recorded using SimpleScreenRecorder demonstrate real-time detection and correct execution of media commands.









Results and Discussion

Experimental results indicate that the system achieves reliable gesture recognition in typical indoor environments with minimal response delay. The use of Media Pipe provides stable hand tracking, while PyAutoGUI ensures compatibility with various media players. Performance degradation is observed under poor lighting or occlusion, highlighting areas for future improvement. Overall, the results validate the feasibility of implementing a practical touchless media control interface on an embedded AI platform

Applications and Future Scope

The proposed system can be deployed in smart homes, classrooms, conference rooms, interactive kiosks, and assistive technologies for users with mobility impairments. Future enhancements include multi-user gesture recognition, adaptive learning of personalized gestures, fusion with voice commands, integration with IoT devices, and improved robustness in complex environments.

Conclusion

This project demonstrates a practical and cost-effective touchless HCI system for media control using hand gestures on the NVIDIA Jetson Nano. The integration of OpenCV, MediaPipe, and PyAutoGUI enables real-time, low-latency interaction without physical contact. The system highlights the potential of embedded AI platforms to support intuitive and accessible human-computer interfaces, paving the way for broader adoption of touchless interaction technologies.

References

- NVIDIA Jetson Nano Developer Guide
- MediaPipe Documentation
- OpenCV Documentation
- PyAutoGUI Documentation
- Research articles on Gesture Recognition and Touchless HCI

Acknowledgements

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